ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2024 P1H Worked Solutions

Nov 2024 | P1H

31 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION Q#: 1 Marks: 5

TOPIC: **Statistics Toolkit**

Nov 2024 P1H - Nov 2024 | P1H

1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	**Frequency**
$10 < p \leqslant 15$	18
$15 < p \leqslant 20$	16
$20 < p \leqslant 25$	14
$25 < p \leqslant 30$	8
$30 < p \leqslant 35$	4

(a) Write down the modal class.

....
(1)

(b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

.... dollars
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 5

TOPIC: **Statistics Toolkit**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

The modal class is the class with the greatest frequency:

$$10 < p \leq 15$$

Use midpoints $12.5, 17.5, 22.5, 27.5, 32.5$.

$$12.5(18) + 17.5(16) + 22.5(14) + 27.5(8) + 32.5(4) = 1170$$

$$\text{estimated mean} = \frac{1170}{60} = 19.5$$

FINAL ANSWER

modal class $10 < p \leq 15$, estimated mean 19.5 dollars.

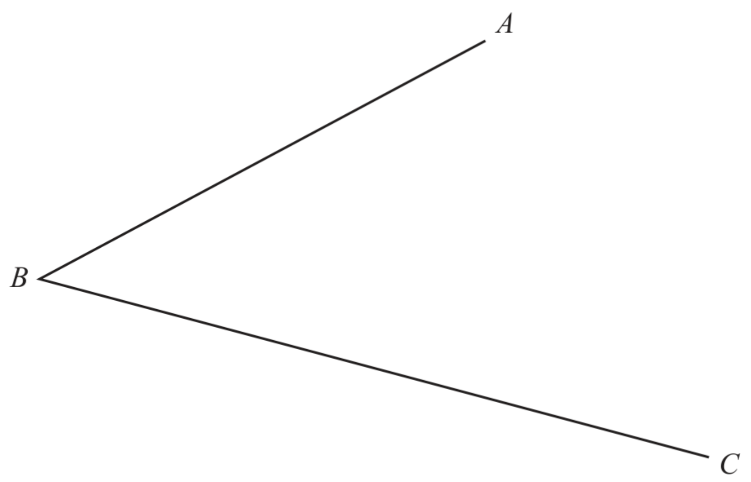
PAST PAPER QUESTION

Q#: 2 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2024 P1H - Nov 2024 | P1H

2 Use ruler and compasses only to construct the bisector of angle ABC
You must show all your construction lines.



(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the bisector of angle ABC:

1. Draw an arc centred at B cutting both arms of the angle.
2. From the two cut points, draw equal arcs inside the angle.
3. Draw the line from B through the intersection of these arcs.

FINAL ANSWER

The constructed ray is the angle bisector of ABC.

PAST PAPER QUESTION

Q#: 3 Marks: 6

TOPIC: **Solving Linear Equations**

Nov 2024 P1H - Nov 2024 | P1H

3 (a) Simplify $\left(p^3\right)^5$

....

(1)

(b) Expand and simplify $2n(4n+3)+n(n-4)$

....

(2)

(c) Solve $\dfrac{2x+5}{3} = 4 - x$

Show clear algebraic working.

$x =$ ....

(3)

(Total for Question 3 is 6 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 6

TOPIC: **Solving Linear Equations**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to solving linear equations. The original tag was expanding brackets, but the equation carries the main marks.

Solution

(a) $(p^3)^5 = p^{15}$.

(b)

$$2n(4n+3) + n(n-4) = 8n^2 + 6n + n^2 - 4n = 9n^2 + 2n$$

(c)

$$\frac{2x+5}{3} = 4 - x$$

$$2x + 5 = 12 - 3x$$

$$5x = 7$$

FINAL ANSWER

(a) p^{15}, (b) $9n^2 + 2n$, (c) $x = \frac{7}{5}$

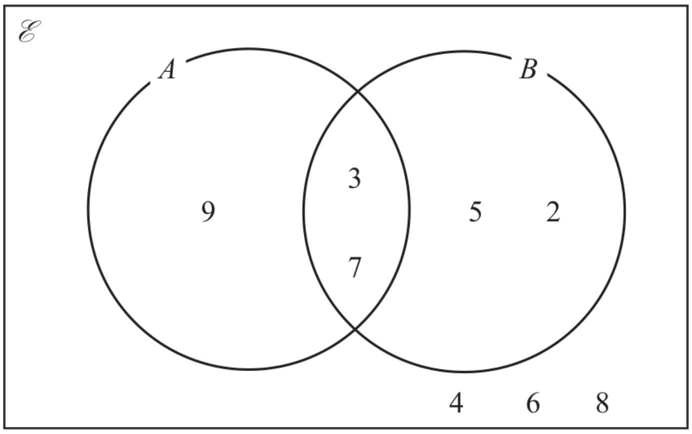
PAST PAPER QUESTION

Q#: 4 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

4 Here is a Venn diagram.



(a) List the members of the set B

....

(1)

(b) List the members of the set $A \cap B$

....

(1)

(c) List the members of the set A'

....

(1)

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

The members of B are all the numbers inside circle B, including the overlap:

$$B = \{2, 3, 5, 7\}$$

The members of $A \cap B$ are the numbers in both circles:

$$A \cap B = \{3, 7\}$$

A' means all the numbers in the universal set that are not in A:

$$A' = \{2, 4, 5, 6, 8\}$$

FINAL ANSWER

(a) $\{2, 3, 5, 7\}$, (b) $\{3, 7\}$, (c) $\{2, 4, 5, 6, 8\}$

PAST PAPER QUESTION

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

Q#: 5 Marks: 4

5 The diagram shows a cylinder.

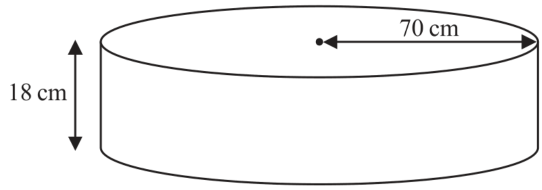

Diagram **NOT** accurately drawn

The radius of the cylinder is 70 cm
The height of the cylinder is 18 cm

Work out the volume of the cylinder.
Give your answer in litres correct to the nearest litre.

.... litres

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

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Topic check. Volume & Surface Area. The tag is correct.

Solution

$$V = \pi r^2 h = \pi(70)^2(18)$$

$$V = 277088.472\ldots\ \text{cm}^3$$

Since $1000\ \text{cm}^3 = 1$ litre,

$$277088.472\ldots\ \text{cm}^3 = 277.088\ldots\ \text{litres}$$

FINAL ANSWER

277 litres.

PAST PAPER QUESTION Q#: 6 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Nov 2024 P1H - Nov 2024 | P1H

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$B = 2^2 \times 5^2 \times 7^2$

$C = 2^2 \times 5^3 \times 7^4$

Find the highest common factor (HCF) of A, B and C
Write your answer as a product of prime factors.

....

(Total for Question 6 is 2 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$A = 2^3 \times 5^4 \times 7 \times 11$$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

The HCF of A, B, and C uses the smallest power of each prime common to all three.

$$2^2, \quad 5^2, \quad 7$$

The prime 11 is not in all three numbers, so it is not included.

$$\text{HCF} = 2^2 \times 5^2 \times 7$$

FINAL ANSWER

$2^2 \times 5^2 \times 7$

PAST PAPER QUESTION

Q#: 7 Marks: 4

TOPIC: **Percentages**

Nov 2024 P1H - Nov 2024 | P1H

7 Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

Shop A	**Shop B**
Our normal price £475 Get 16% off our normal price	Get 15% off our normal price Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Percentages**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Corrected to Percentages. This is a percentage discount comparison.

Method

Shop A discount:

$$16\% \text{ of } 475 = 0.16 \times 475 = 76$$

Shop B: 408 euros is 85% of the normal price.

$$\text{normal price} = 408 \div 0.85 = 480$$

Shop B discount:

$$480 - 408 = 72$$

Since

$$76 > 72$$

Shop A gives the greater discount.

FINAL ANSWER

Shop A gives the greater discount.

PAST PAPER QUESTION

Q#: 8 Marks: 6

TOPIC: **Factorising**

Nov 2024 P1H - Nov 2024 | P1H

8 (a) (i) Factorise $x^2 + 5x - 24$

....

(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

....

(1)

(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

....

(3)

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 6

TOPIC: **Factorising**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is acceptable because the quadratic factorising section is a full half of the question.

Solution

(a)(i)

$$x^2 + 5x - 24 = (x + 8)(x - 3)$$

(a)(ii) $x = -8$ or $x = 3$.

(b)

$$3y + 5 > 7y - 10$$

$$15 > 4y$$

$$y < \frac{15}{4}$$

FINAL ANSWER

$(x + 8)(x - 3)$, $x = -8$ or $x = 3$, $y < \frac{15}{4}$

PAST PAPER QUESTION Q#: 9 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Nov 2024 P1H - Nov 2024 | P1H

9 (a) Write 8.4×10^{-5} as an ordinary number.

....

(1)

(b) Work out $\left(6.5 \times 10^{-40}\right) \times \left(8 \times 10^{185}\right)$

Give your answer in standard form.

....

(2)

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$8.4 \times 10^{-5} = 0.000084$$

For part (b),

$$(6.5 \times 10^{-40})(8 \times 10^{185}) = 52 \times 10^{145}$$

Write this in standard form:

$$52 \times 10^{145} = 5.2 \times 10^{146}$$

FINAL ANSWER

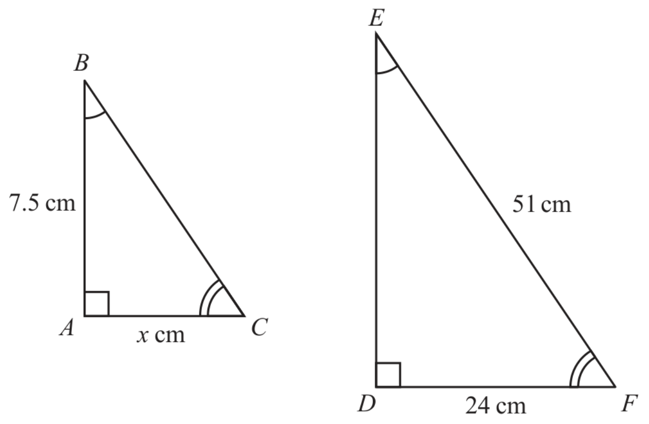
(a) 0.000084, (b) 5.2×10^{146}

PAST PAPER QUESTION Q#: 10 Marks: 5

TOPIC: **Congruence, Similarity & Geometrical Proof**

Nov 2024 P1H - Nov 2024 | P1H

10 Here are two similar triangles.



Diagrams **NOT** accurately drawn

Work out the value of x
Show your working clearly.

$x =$ ....

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Congruence, Similarity & Geometrical Proof**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Congruence, Similarity & Geometrical Proof. The tag is correct.

Solution

In the larger triangle,

$$DE^2 = 51^2 - 24^2$$

$$DE = 45$$

The scale factor from the large triangle to the small triangle is

$$\frac{7.5}{45} = \frac{1}{6}$$

So

$$x = 24 \times \frac{1}{6} = 4$$

FINAL ANSWER

$x = 4$.

PAST PAPER QUESTION Q#: 11 Marks: 4

TOPIC: **Graphs of Functions**

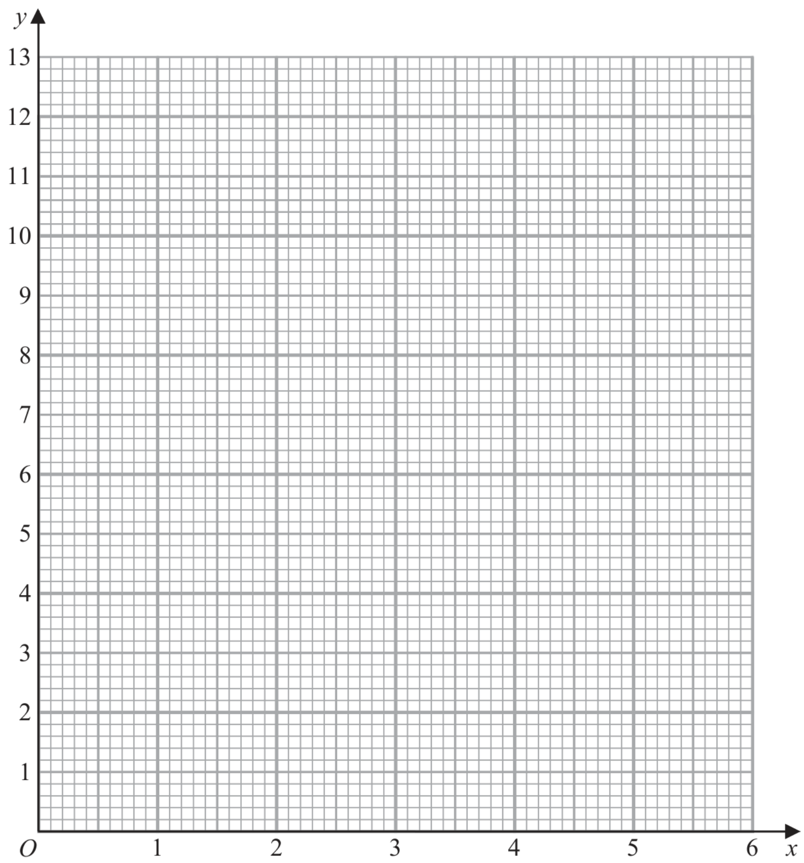
Nov 2024 P1H - Nov 2024 | P1H

11 (a) Complete the table of values for $y = 2\left(x + \dfrac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

(b) On the grid, draw the graph of $y = 2\left(x + \dfrac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 6$



(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION Q#: 11 Marks: 4

TOPIC: **Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Correct. This is a reciprocal-curve table and graph question.

Solution

$$y = 2\left(x + \frac{1}{x}\right)$$

The completed table is:

x	0.5	1	2	3	4	5	6
y	5	4	5	6.7	8.5	10.4	12.3

Plot the points and join them with a smooth curve.

FINAL ANSWER

Missing values are $5, 4, 8.5, 10.4$.

PAST PAPER QUESTION

Q#: 12 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P1H - Nov 2024 | P1H

12 ABC is a right-angled triangle.
D is a point on BC

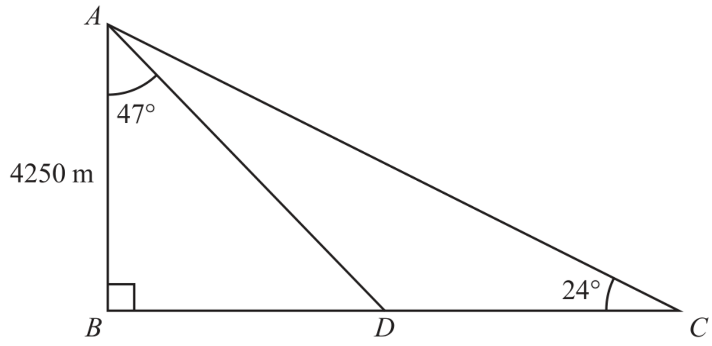


Diagram **NOT** accurately drawn

$AB = 4250$ m angle $BAD = 47°$ angle $BCA = 24°$

Work out the length of DC
Give your answer correct to the nearest integer.

.... m

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

In triangle ABD,

$$\tan 47^\circ = \frac{BD}{4250}$$

$$BD = 4250 \tan 47^\circ = 4557.567\ldots$$

In triangle ABC,

$$\tan 24^\circ = \frac{4250}{BC}$$

$$BC = \frac{4250}{\tan 24^\circ} = 9545.656\ldots$$

$$DC = BC - BD = 4988.089\ldots$$

FINAL ANSWER

4988 m.

PAST PAPER QUESTION Q#: 13 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

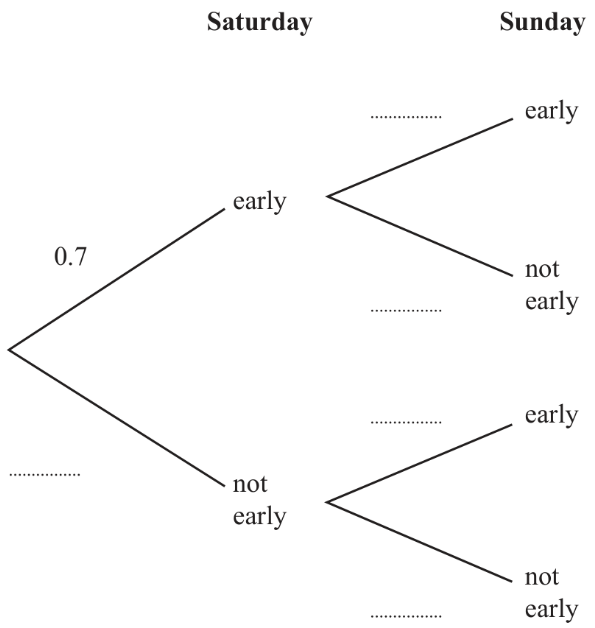
Nov 2024 P1H - Nov 2024 | P1H

13 The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

(a) Use this information to complete the probability tree diagram.



(2)

(b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

....

(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

$$P(\text{not early on Saturday}) = 1 - 0.7 = 0.3$$

$$P(\text{early on both days}) = 0.7 \times 0.9 = 0.63$$

FINAL ANSWER

0.63.

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

Nov 2024 P1H - Nov 2024 | P1H

14 B is inversely proportional to the square of d

$B = 0.25$ when $d = 12$

Find a formula for B in terms of d

....

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Direct and Inverse Proportion. The tag is correct.

Method

B is inversely proportional to d^2, so

$$B = \frac{k}{d^2}$$

Use $B = 0.25$ when $d = 12$:

$$0.25 = \frac{k}{12^2}$$

$$k = 0.25 \times 144 = 36$$

Therefore

$$B = \frac{36}{d^2}$$

FINAL ANSWER

$B = \frac{36}{d^2}$.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2024 P1H - Nov 2024 | P1H

15 Write $3x(2x-1)(5x+4)$ in the form ax^3+bx^2+cx where a, b and c are integers.

....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Expanding Brackets. This is an expansion question, not a surds question.

Method

$$3x(2x-1)(5x+4)$$

First expand the two brackets:

$$(2x-1)(5x+4) = 10x^2 + 8x - 5x - 4$$

$$= 10x^2 + 3x - 4$$

Now multiply by $3x$:

$$3x(10x^2 + 3x - 4) = 30x^3 + 9x^2 - 12x$$

FINAL ANSWER

$30x^3 + 9x^2 - 12x$

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2024 P1H - Nov 2024 | P1H

16 *OAPB* is a sector of a circle, centre *O*

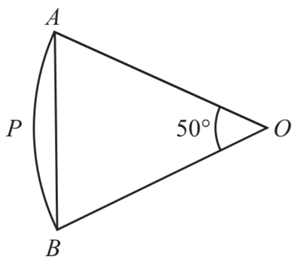


Diagram **NOT** accurately drawn

Angle $AOB = 50°$

Area of triangle $OAB = 120\text{ cm}^2$

Work out the area of the sector *OAPB*
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Circles, Arcs & Sectors.

Solution

For triangle OAB,

$$120 = \frac{1}{2}r^2 \sin 50^\circ$$

$$r^2 = \frac{240}{\sin 50^\circ}$$

Area of sector $OAPB$:

$$\frac{50}{360}\pi r^2 = \frac{50}{360}\pi\left(\frac{240}{\sin 50^\circ}\right) = 136.701\ldots$$

FINAL ANSWER

137 cm^2.

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Algebraic Proof**

Nov 2024 P1H - Nov 2024 | P1H

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

....

(1)

(b) Show that $\dfrac{5-\sqrt{2}}{\sqrt{2}-1}$ can be written in the form $a+b\sqrt{2}$ where a and b are integers.

(3)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Algebraic Proof**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

Solution

(a)

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}$$

So $n = 5$.

(b)

$$\frac{5-\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = \frac{(5-\sqrt{2})(\sqrt{2}+1)}{2-1}$$

$$= 5\sqrt{2} + 5 - 2 - \sqrt{2}$$

$$= 3 + 4\sqrt{2}$$

FINAL ANSWER

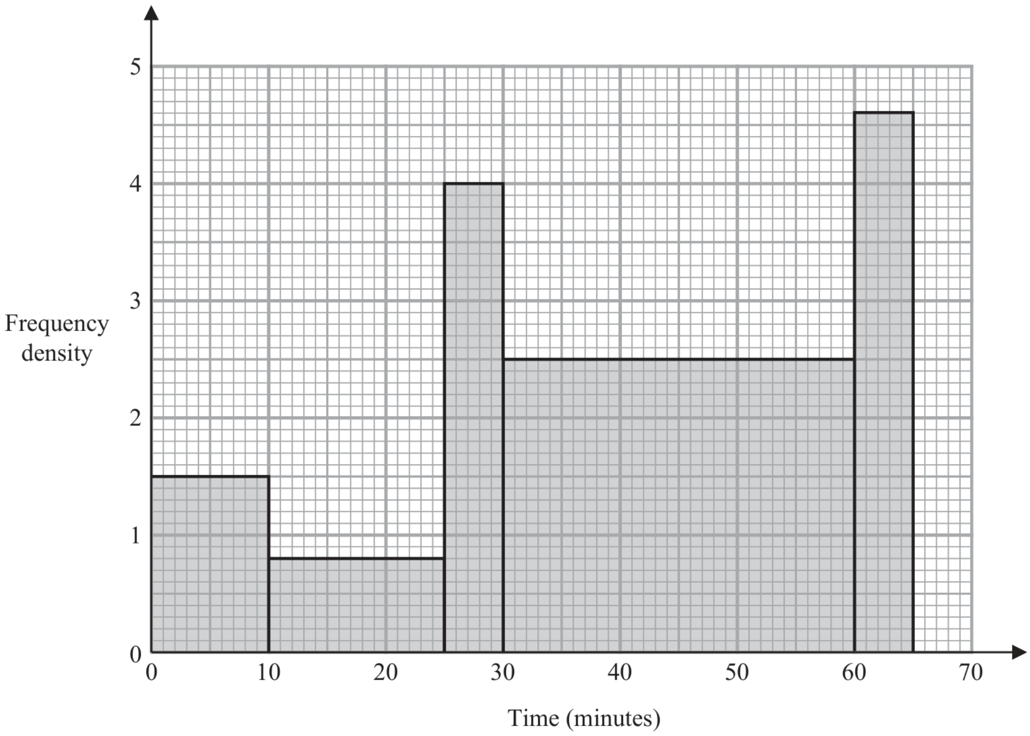
$n = 5$, and $3 + 4\sqrt{2}$.

PAST PAPER QUESTION Q#: 18 Marks: 4

TOPIC: **Histograms**

Nov 2024 P1H - Nov 2024 | P1H

18 The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

....

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Histograms**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Histograms. The tag is correct.

Solution

Estimate the frequencies from each bar:

$$0 \text{ to } 10: \ 10(1.5) = 15$$

$$10 \text{ to } 25: \ 15(0.8) = 12$$

$$25 \text{ to } 30: \ 5(4) = 20$$

$$30 \text{ to } 60: \ 30(2.5) = 75$$

$$60 \text{ to } 65: \ 5(4.6) = 23$$

Total frequency:

$$15 + 12 + 20 + 75 + 23 = 145$$

Between 20 and 60 minutes:

$$5(0.8) + 20 + 75 = 99$$

$$\text{fraction} = \frac{99}{145}$$

FINAL ANSWER

$\frac{99}{145}$.

PAST PAPER QUESTION

Q#: 19 Marks: 5

Topic: **Sequences**

Nov 2024 P1H - Nov 2024 | P1H

19 An arithmetic series has first term *a* and common difference *d*

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

....

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Sequences**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Sequences. The tag is correct.

Solution

Let the first term be a and the common difference be d.

$$S_{30} = 4395$$

$$\frac{30}{2}(2a + 29d) = 4395$$

$$2a + 29d = 293$$

The 8th term plus the 20th term is 284:

$$(a + 7d) + (a + 19d) = 284$$

$$2a + 26d = 284$$

Subtract the equations:

$$3d = 9$$

$$d = 3$$

Then

$$2a + 26(3) = 284$$

$$2a = 206$$

$$a = 103$$

Now find S_{45}:

$$S_{45} = \frac{45}{2}(2a + 44d)$$

$$S_{45} = \frac{45}{2}(206 + 132) = 7605$$

FINAL ANSWER

7605.

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

....

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

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Topic check. Moved from Estimating Gradients to Differentiation.

Method

$$y = 2x^4 - 64x$$

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

Find the y-coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

FINAL ANSWER

$y = -96$

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

....

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Estimating Gradients to Differentiation.

Method

$$y = 2x^4 - 64x$$

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

Find the y-coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

FINAL ANSWER

$y = -96$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

21 $T = \dfrac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

$$T = \frac{x^2 + y^2}{w}$$

For the upper bound, use the upper bounds of x and y, and the lower bound of w.

$$x < 28.45, \quad y < 17.5, \quad w \geq 87.5$$

$$T_{\text{upper}} = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\ldots$$

Correct to 3 significant figures,

$$12.7503\ldots \approx 12.8$$

FINAL ANSWER

12.8

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

21 $T = \dfrac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation & Bounds is correct.

Method

$$T = \frac{x^2 + y^2}{w}$$

For the upper bound of T, use the upper bounds of x and y, and the lower bound of w.
$x = 28.45, \quad y = 17.5, \quad w = 87.5$

$$T = \frac{28.45^2 + 17.5^2}{87.5}$$

$$T = 12.7503...$$

FINAL ANSWER
12.8 to 3 significant figures.

PAST PAPER QUESTION

Q#: 22 **Marks: 3**

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

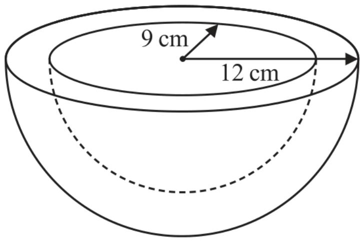


Diagram **NOT** accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

.... cm^3

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The bowl is a hemispherical shell.
The volume of the metal is:

$$\text{outer hemisphere volume} - \text{inner hemisphere volume}$$

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

So

$$\text{metal volume} = \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3$$

$$= \frac{2}{3}\pi(12^3 - 9^3)$$

$$= \frac{2}{3}\pi(1728 - 729)$$

$$= 666\pi$$

$$= 2092.3007\ldots$$

FINAL ANSWER

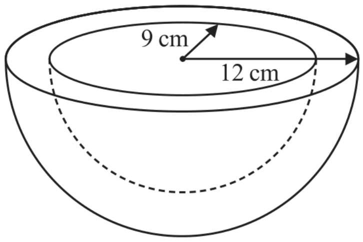
$\boxed{2092 \text{ cm}^3}$

PAST PAPER QUESTION Q#: 22 Marks: 3

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.



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Work out the volume of the metal.
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.... cm^3

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The bowl is a hemispherical shell.
The volume of the metal is:

$$\text{outer hemisphere volume} - \text{inner hemisphere volume}$$

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

So

$$\text{metal volume} = \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3$$

$$= \frac{2}{3}\pi(12^3 - 9^3)$$

$$= \frac{2}{3}\pi(1728 - 729)$$

$$= 666\pi$$

$$= 2092.3007\ldots$$

FINAL ANSWER

$\boxed{2092 \text{ cm}^3}$

PAST PAPER QUESTION Q#: 23 Marks: 5

TOPIC: **Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

$k =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Graphs of Functions.

Method

The curve is

$$y = x^2 - 8x - 9$$

The line is

$$y = k$$

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

The two roots are p and q.
For this quadratic,

$$p + q = 8$$

We are also given

$$p - q = 14$$

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

Use one root in $y = x^2 - 8x - 9$:

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

FINAL ANSWER

$k = 24$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

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The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

$k =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Graphs of Functions.

Method

The curve is

$$y = x^2 - 8x - 9$$

The line is

$$y = k$$

At the intersection points,

$$x^2 - 8x - 9 = k$$

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$$k = 24$$

FINAL ANSWER

$k = 24$

PAST PAPER QUESTION

TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

Q#: 24 Marks: 6

24 OAB is a triangle.

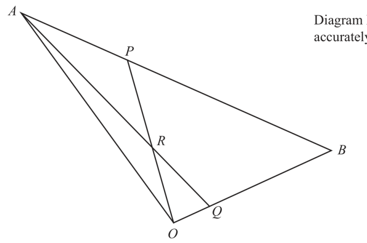

Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 10\mathbf{a}$ $\overrightarrow{OB} = 10\mathbf{b}$

ARQ and ORP are straight lines.

$\overrightarrow{AP} = \frac{1}{4}\overrightarrow{AB}$ and $\overrightarrow{OQ} = \frac{1}{5}\overrightarrow{OB}$

Write the following vectors in terms of **a** and **b**
Simplify your answers.

(i) $\overrightarrow{AQ}$

....
(1)

(ii) $\overrightarrow{OP}$

....
(1)

(iii) $\overrightarrow{OR}$

....
(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 6

TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 10\mathbf{a}, \qquad \vec{OB} = 10\mathbf{b}$$

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

For part (iii), R lies on OP, so

$$\vec{OR} = t\vec{OP}$$

It also lies on AQ, so

$$\vec{OR} = \vec{OA} + s\vec{AQ}$$

$$\vec{OR} = 10\mathbf{a} + s(2\mathbf{b} - 10\mathbf{a})$$

$$\vec{OR} = (10 - 10s)\mathbf{a} + 2s\mathbf{b}$$

Equate with

$$t\left(\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}\right)$$

$$\frac{15}{2}t = 10 - 10s$$

$$\frac{5}{2}t = 2s$$

From the second equation,

$$s = \frac{5}{4}t$$

Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

FINAL ANSWER

$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$, $\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$, $\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$

PAST PAPER QUESTION

Q#: 24 Marks: 6

Topic: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

24 OAB is a triangle.

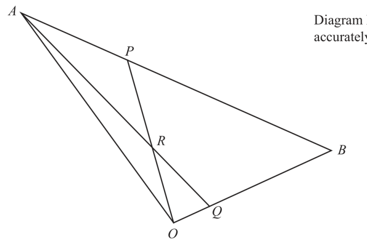

Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 10\mathbf{a}$ $\quad \overrightarrow{OB} = 10\mathbf{b}$

ARQ and ORP are straight lines.

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Write the following vectors in terms of **a** and **b**
Simplify your answers.

(i) $\overrightarrow{AQ}$

....

(1)

(ii) $\overrightarrow{OP}$

....

(1)

(iii) $\overrightarrow{OR}$

....

(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Linear Graphs to Vectors.

Method

$$\vec{OA} = 10\mathbf{a}, \qquad \vec{OB} = 10\mathbf{b}$$

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

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$$\vec{OR} = t\vec{OP}$$

It also lies on AQ, so

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Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

FINAL ANSWER
$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$, $\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$, $\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$

PAST PAPER QUESTION

Q#: 25 Marks: 4

TOPIC: **Functions**

Nov 2024 P1H - Nov 2024 | P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geqslant 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) =$ ....

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 4

TOPIC: **Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = 2x^2 - 24x + 7$$

Complete the square:

$$f(x) = 2(x^2 - 12x) + 7$$

$$f(x) = 2[(x-6)^2 - 36] + 7$$

$$f(x) = 2(x-6)^2 - 65$$

Let $y = f(x)$.

$$y = 2(x-6)^2 - 65$$

$$y + 65 = 2(x-6)^2$$

$$\frac{y+65}{2} = (x-6)^2$$

Since $x \geq 6$, take the positive square root:

$$x - 6 = \sqrt{\frac{y+65}{2}}$$

$$x = 6 + \sqrt{\frac{y+65}{2}}$$

Replace y by x.

FINAL ANSWER

$f^{-1}(x) = 6 + \sqrt{\dfrac{x+65}{2}}$

PAST PAPER QUESTION

Q#: 25 Marks: 4

TOPIC: **Functions**

Nov 2024 P1H - Nov 2024 | P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geqslant 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) =$ ....

(Total for Question 25 is 4 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 4

TOPIC: **Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = 2x^2 - 24x + 7$$

Complete the square:

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Replace y by x.

FINAL ANSWER

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